Supplemental Appendix

A Khan and Thomas (2008) Model

This appendix evaluates the precomputation method of Section 3 in the model of Khan and Thomas (2008). Unlike the Bloom et al. (2018) implementation, the price search in this exercise does not interpolate excess demand. The comparison between the error-price and raw-price formulations therefore shows directly that centering the price grid on the forecast can substantially affect accuracy even when both formulations use the same number of price nodes.

A.1 Model

The Khan and Thomas (2008) model used here contains aggregate and idiosyncratic productivity shocks but no stochastic-volatility uncertainty shock. The household block is the same as in the baseline Bloom et al. (2018) model, so prices satisfy (8). Firms produce according to $y = z\varepsilon k^{\alpha} n^{\nu}$. Labor is chosen frictionlessly each period, whereas changing capital from its depreciated level $(1-\delta)k$ requires paying the stochastic fixed cost $\xi\omega$, where ξ is drawn independently each period from a distribution G with support $[0, \bar{\xi}]$.[13] Let μ denote the distribution of firms over (ε, k). Measuring the firm value in units of household marginal utility, $V \equiv pv$, the Bellman equation is

$$V^1(\varepsilon, k, \xi; z, \mu) = \max_{n \in \mathbf{R}_+} (z\varepsilon k^{\alpha} n^{\nu} - \omega n + (1-\delta)k)p + \max\left\{ -\xi\omega p + \max_{k^* \in \mathbf{R}_+} R(\varepsilon, k^*; z, \mu), R(\varepsilon, (1-\delta)k; z, \mu) \right\} \tag{36}$$

$$R(\varepsilon, k'; z, \mu) \equiv -k'p + \beta \mathbb{E}_{\varepsilon, z, \mu}\left[V^0(\varepsilon', k'; z', \mu') \right],$$

$$V^0(\varepsilon, k; z, \mu) \equiv \int_0^{\bar{\xi}} V^1(\varepsilon, k, \xi; z, \mu) G(d\xi)$$

Forecasting rules,

$$\mu' = \Gamma_{\mu}(z, \mu), \quad p = \Gamma_p(z, \mu)$$

Here R is the continuation return in marginal-utility units. Note that the second maximization in (36) reflects the firm's choice between investing to move to the new capital level k^* or remaining at the depreciated level $(1-\delta)k$. Importantly, the choice of k^* depends only on (z, ε, μ) and **not** on the firm's individual capital k.[14] The threshold level of the adjustment cost ξ that determines

[13] For comparability with Terry (2017), I omit the zero-investment band in the original Khan and Thomas (2008) specification.

[14] Since the adjustment cost $\xi\omega p$ is independent of k, the optimal choice of k^* is also independent of it.

Table 6: Computation time in the Khan–Thomas model

Model	Method	Total	VFI	Precompute policy	Simulation
KT	Baseline	16 min	3 sec	–	180 sec
KT	Proposed	80 sec	3 sec	3 sec	10 sec

whether the firm invests is given by:

$$\begin{aligned} \widehat{\xi}(\varepsilon, k; z, \mu) &= \frac{R(\varepsilon, k^*; z, \mu) - R(\varepsilon, (1-\delta)k; z, \mu)}{p\omega}, \\ \xi^*(\varepsilon, k; z, \mu) &= \min\{\bar{\xi}, \max\{0, \widehat{\xi}(\varepsilon, k; z, \mu)\}\}. \end{aligned} \tag{37}$$

The cutoff is restricted to the support of the cost distribution. For a continuous G, the choice at the cutoff does not affect aggregate quantities.

A.2 Precomputation: Computation Time and Accuracy

Speed comparison Table 6 reports the total computation time and its main components: the value-function iteration (VFI), the policy precomputation step, and the simulation step.

The proposed method reduces total computation time from 16 minutes to 80 seconds, corresponding to roughly a twelve-fold speedup. The value-function iteration itself takes 3 seconds in both methods, so the improvement does not come from solving the Bellman equation faster. Instead, the main driver of the speed gain is the simulation step. [15]

Accuracy check. I next examine whether this acceleration preserves the simulated dynamics of the baseline KS method. Figure 7 compares the simulation paths generated by the baseline KS method, the error-price formulation, and the raw-price formulation over periods 1500–1600. The error-price series is almost indistinguishable from the baseline KS path, indicating that the proposed formulation reproduces the benchmark dynamics with essentially no visible discrepancy. By contrast, the raw-price formulation displays noticeable deviations from the baseline path.

I further compare accuracy using investment statistics computed from the simulated paths. Table 7 reports moments of the investment rate i/k for the baseline KS method, the error-price formulation, and the raw-price formulation.

[15] In terms of computation time, there is essentially no difference between the raw-price formulation and the error-price formulation; both formulations deliver the same speed improvement because they eliminate the repeated maximization inside the price-search loop.

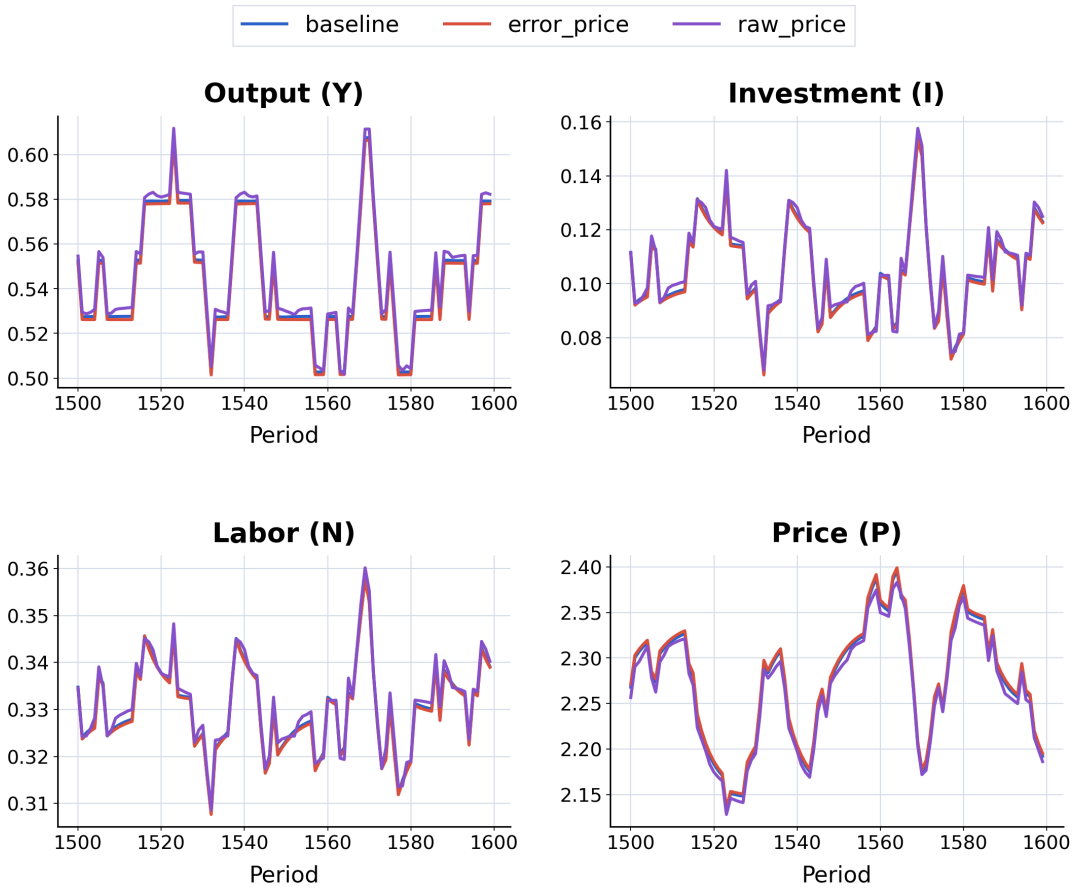


Figure 7: Simulation path: baseline KS, error-price, and raw-price formulations

Table 7: Investment statistics in the Khan–Thomas model

Moment	KS	Error price	Raw price
Mean i/k	0.09487	0.09488	0.09515
Std. i/k	0.25973	0.25982	0.25625
Pr(inaction)	0.76882	0.76764	**0.64339**
$\Pr(i/k \geq 0.2)$	0.17271	0.17266	**0.18483**
$\Pr(i/k \leq -0.2)$	0.02806	0.02804	**0.04364**
$\Pr(i/k > 0)$	0.18948	0.19015	**0.27216**
$\Pr(i/k < 0)$	0.04171	0.04221	**0.08445**

The investment statistics reinforce the evidence from the simulated paths. The error-price formulation matches the KS benchmark almost exactly across all reported moments. The mean and standard deviation of i/k are virtually unchanged, and the probabilities of inaction, positive investment, and negative investment remain extremely close to their KS values. By contrast, the raw-price formulation generates sizable discrepancies, especially along the extensive margin. The inaction probability falls from 0.76882 in the KS benchmark to 0.64339 under the raw-price formulation, while the probabilities of positive and negative investment rise substantially.

B Additional Results of the NN-KS method for Bloom et al. (2018) Model

B.1 Hyperparameters and numerical settings

Table 8: NN–KS hyperparameters and numerical settings

Component	Setting	Value
Value network	Inputs; output	$(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K)$; scalar $\tilde{V}_{\text{nn}}$
	Hidden layers; activation	$256 \times 256 \times 256$; ReLU
Discrete gate	Inputs; output	seven states plus p_{error}; four regime logits
	Hidden layers; activation	$256 \times 256 \times 256$; ReLU
Conditional policy	Inputs; output	seven states, p_{error}, two-bit r; (k', n)
	Hidden layers; activation	$256 \times 256 \times 256$; ReLU
Optimization	Optimizer; learning-rate schedule	Adam; 10^{-4} toward 5×10^{-5}
	Main samples per component	500,000
	Main batch size; epochs	256; value 2, policy 1, gate 1
	Target-network update	$\tau = 0.05$
State grids	$(z, \varepsilon, k, n_{-1}, \sigma, \sigma_{-1}, K)$	$(5, 5, 91, 37, 2, 2, 10)$; 3,367,000 states
Price error	Input normalization	$p_{\text{error}}/0.05$ before network evaluation
	Ordinary additive perturbation	$[-0.05, 0.05]$
Simulation	Length; burn-in; price iterations	5,000; 500; at most 20
Precision	Networks; aggregation/distribution	FP32; FP64

B.2 Results

B.2.1 Bellman Errors: KS and NN–KS

This subsection reports relative Bellman errors for the saved KS and NN–KS value functions. Retain $x_f = (k, n_{-1}, \varepsilon)$, $a = (k', n)$, and $k_0 = (1 - \delta_k)k$ from Section 3. Let $\tilde{V}_m(x_f; Z, K)$ denote the saved value function for method $m \in \{\text{KS}, \text{NN–KS}\}$, and let θ_m denote its associated forecasting-rule parameters. Define

$$\begin{aligned}(T_{\theta_m}\tilde{V}_m)(x_f; Z, K) = \max_{a \in \mathcal{A}(x_f, Z)} \Big\{ \widehat{p}_m \left[y - \widehat{w}_m n - (k' - k_0) - \text{AC}^k - \text{AC}^n(\widehat{w}_m) \right] \\ + \beta \mathbb{E}\left[\tilde{V}_m\left(k', n, \varepsilon'; Z', \Gamma_K(Z, K; \theta_m)\right) \right] \Big\},\end{aligned} \tag{38}$$

where $\widehat{p}_m = \Gamma_p(Z, K; \theta_m)$ and $\widehat{w}_m = \chi / \widehat{p}_m$, as in (15). This is the same firm problem as in (16), evaluated using each method's value function and forecasting rules of the final outer loop.

I evaluate the Bellman operator itself rather than substituting the saved policy. The KS solution is computed on a discrete choice grid, so multilinear interpolation of $\tilde{V}_m$ is used to evaluate the continuation value at continuous choices. The NN–KS value network is already continuous in (k', n). For both methods, I maximize the right-hand side over $\mathcal{A}(x_f, Z)$ and use the resulting maximized value to construct the residual. The expectation is taken over all next-period shock realizations. The relative Bellman error is

$$B_m(x_f; Z, K) = \frac{\left| (T_{\theta_m}\tilde{V}_m)(x_f; Z, K) - \tilde{V}_m(x_f; Z, K) \right|}{\max\left\{ \left| \tilde{V}_m(x_f; Z, K) \right|, 10^{-10} \right\}}. \tag{39}$$

I compute the mean of $B_m(x_f; Z, K)$ at 1,000 common grid nodes and 1,000 common off-grid states.

Table 9: Mean relative Bellman errors in the baseline model

Evaluation states	KS	NN–KS
Grid nodes	3.68667×10^{-7}	3.21681×10^{-4}
Off-grid states	5.44973×10^{-6}	3.22607×10^{-4}

Notes: Entries report $B_m(x_f; Z, K)$ as a ratio, not a percentage. Each evaluation sample contains 1,000 states shared by both methods. The NN–KS entries average the mean errors from five independent runs.

The NN–KS mean error is approximately 3.22×10^{-4}, or 0.032%, at both grid and off-grid states.

B.2.2 Shock-specific forecasting accuracy

Table 10 conditions forecasting accuracy on the aggregate state (z, σ, σ_{-1}).

Table 10: Bloom shock-specific forecast accuracy

Aggregate state	Price p				Capital K			
(z, σ, σ_{-1})	KS		NN–KS		KS		NN–KS	
	RMSE (%)	R^2	RMSE (%)	R^2	RMSE (%)	R^2	RMSE (%)	R^2
				Den Haan statistics				
Maximum	3.80		3.59		5.68		5.77	
Mean	0.96		0.82		1.83		1.81	
(1,0,0)	0.61	0.76	0.58	0.82	0.33	0.98	0.37	0.99
(1,0,1)	0.10	0.99	0.20	0.96	0.05	1.00	0.12	1.00
(1,1,0)	0.19	0.97	0.23	0.96	0.52	0.95	0.59	0.94
(1,1,1)	0.29	0.91	0.28	0.95	0.11	1.00	0.15	1.00
(2,0,0)	0.51	0.81	0.50	0.86	0.34	0.98	0.35	0.99
(2,0,1)	0.13	0.98	0.14	0.99	0.08	1.00	0.08	1.00
(2,1,0)	0.39	0.84	0.41	0.84	0.27	0.97	0.27	0.98
(2,1,1)	0.33	0.93	0.38	0.94	0.11	1.00	0.16	1.00
(3,0,0)	0.43	0.92	0.46	0.92	0.28	0.99	0.27	0.99
(3,0,1)	0.19	0.99	0.23	0.98	0.10	1.00	0.11	1.00
(3,1,0)	0.23	0.98	0.29	0.97	0.37	0.98	0.45	0.99
(3,1,1)	0.37	0.95	0.41	0.93	0.12	1.00	0.14	1.00
(4,0,0)	0.49	0.93	0.54	0.86	0.32	0.99	0.31	0.99
(4,0,1)	0.17	0.99	0.22	0.97	0.09	1.00	0.12	1.00
(4,1,0)	0.46	0.95	0.31	0.93	0.39	0.98	0.40	0.98
(4,1,1)	0.36	0.94	0.44	0.88	0.12	1.00	0.13	1.00
(5,0,0)	0.53	0.91	0.51	0.89	0.33	0.99	0.30	0.99
(5,0,1)	0.11	0.99	0.22	0.96	0.22	0.99	0.16	1.00
(5,1,0)	0.33	0.98	0.34	0.93	0.13	1.00	0.12	1.00
(5,1,1)	0.31	0.97	0.31	0.93	0.12	1.00	0.12	1.00

Note: The table reports statistics conditional on $(z, \sigma, \sigma_{t-1})$. RMSE and Den Haan entries are percentage points of log deviations (log errors multiplied by 100).

C Model Details and Additional Results for Section 5

This appendix describes how the household-preference extension in Section 5 is solved with NN–KS and provides additional quantitative results. It then varies the Frisch elasticity and consumption–hours complementarity.

C.1 Model and NN–KS Implementation

The firm side of the model is unchanged from Section 2. A firm has the individual state $x_f = (k, n_{-1}, \varepsilon)$, produces $y = z\varepsilon k^{\alpha} n^{\nu}$, and chooses next-period capital k' and current employment n subject to the capital and labor adjustment costs described in Section 2.1. The preference extension replaces the baseline household utility function with a general period utility $U(C, N)$. The two household price conditions are

$$p = U_C(C, N), \qquad w = -\frac{U_N(C, N)}{U_C(C, N)}. \tag{40}$$

Unlike in the baseline model, p and w cannot generally be recovered from one another. Consequently, the simulation searches jointly for both prices. For each candidate pair (p, w), the firm policies imply aggregate employment $N(p, w)$ and, through the resource constraint, consumption $C(p, w)$. The two market-clearing residuals are

$$p - U_C\big(C(p,w), N(p,w)\big) = 0, \qquad w + \frac{U_N\big(C(p,w), N(p,w)\big)}{U_C\big(C(p,w), N(p,w)\big)} = 0. \tag{41}$$

To approximate the distributional state, I retain aggregate capital K_t and add lagged aggregate employment N_{t-1} as a second moment. The latter is useful to forecast the wage. With $Z_t = (z_t, \sigma_t, \sigma_{t-1})$, NN–KS therefore uses three forecasting rules:

$$K_{t+1} = \Gamma_K(Z_t, K_t, N_{t-1}; \theta), \tag{42}$$

$$\widehat{p}_t = \Gamma_p(Z_t, K_t, N_{t-1}; \theta), \tag{43}$$

$$\widehat{w}_t = \Gamma_w(Z_t, K_t, N_{t-1}; \theta). \tag{44}$$

To evaluate the continuation value, current aggregate employment N_t, which becomes the lagged-employment moment next period, is backed out from the household wage condition at $(\widehat{p}_t, \widehat{w}_t)$. For example, under the separable finite-Frisch specification,

$$p_t w_t = \chi N_t^{\phi} \quad \Longrightarrow \quad \widehat{N}_t = \left(\frac{\widehat{p}_t \widehat{w}_t}{\chi}\right)^{1/\phi}, \qquad \phi > 0. \tag{45}$$

For the other preference specifications, $\widehat{N}_t$ is obtained from the corresponding household price

equations in (40). Thus, the continuation value at a candidate firm action uses

$$\widetilde{V}_\theta\left(k', n, \varepsilon'; Z', \Gamma_K(Z_t, K_t, N_{t-1}; \theta), \widehat{N}_t\right). \tag{46}$$

Relative to the implementation in Section 4, the value network therefore gains lagged aggregate employment as an input:

$$\widetilde{V}_{\text{nn}} = \widetilde{V}_{\text{nn}}(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K, N_{-1}). \tag{47}$$

The discrete and conditional policy networks additionally take the candidate wage. In the implementation, the two candidate prices are represented as additive deviations from their respective forecasts,

$$p = \Gamma_p(Z, K, N_{-1}; \theta) + p_{\text{error}}, \qquad w = \Gamma_w(Z, K, N_{-1}; \theta) + w_{\text{error}}. \tag{48}$$

Hence, their common economic inputs are

$$(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K, N_{-1}, p_{\text{error}}, w_{\text{error}}), \tag{49}$$

with the adjustment-regime indicators also supplied to the conditional policy network. During policy training, p_{error} and w_{error} are sampled independently from the interval $[-0.05, 0.05]$, even though current aggregate labor N for the continuation value is computed by the forecasting rule with $p_{\text{error}} = w_{\text{error}} = 0$. Value-network updates use the forecast pair, $p_{\text{error}} = w_{\text{error}} = 0$.

C.2 Sensitivity to the Frisch Elasticity

This subsection examines how the finite Frisch elasticity affects the wage channel in Section 5.1 and explains the choice of a Frisch elasticity of 2 in the main analysis. I compare separable preferences with elasticities $1/\phi \in \{0.5, 1, 2\}$, keeping log utility from consumption. For each elasticity, the labor-disutility weight is recalibrated to the common reference allocation $(\bar{C}, \bar{N}, \bar{w}) = (0.73, 0.299, 1.451)$; the remaining structural parameters are unchanged.

Uncertainty-shock responses. Figure 8 shows that a smaller Frisch elasticity produces a larger wage decline and a smaller employment contraction. The household condition in equation (29) implies

$$\Delta \log w_t = \Delta \log C_t + \phi \, \Delta \log N_t. \tag{50}$$

A lower Frisch elasticity corresponds to a larger ϕ, so a given decline in employment exerts more downward pressure on the wage. In equilibrium, this wage adjustment cushions the fall in firms' labor demand. As the Frisch elasticity falls from 2 to 1 and 0.5, the impact wage decline increases from 2.54% to 3.16% and 4.52%, while the employment decline falls from 4.33% to 2.73% and 2.07%.

The smaller employment contraction also supports investment through capital–labor complementarity. By preserving employment, the larger wage decline cushions the fall in the marginal product of capital and reduces the incentive to postpone capital investment. The impact investment decline is 6.81% at a Frisch elasticity of 2, compared with 3.06% at 1 and 1.54% at 0.5. Output contracts less as well, while consumption falls on impact in all three cases.

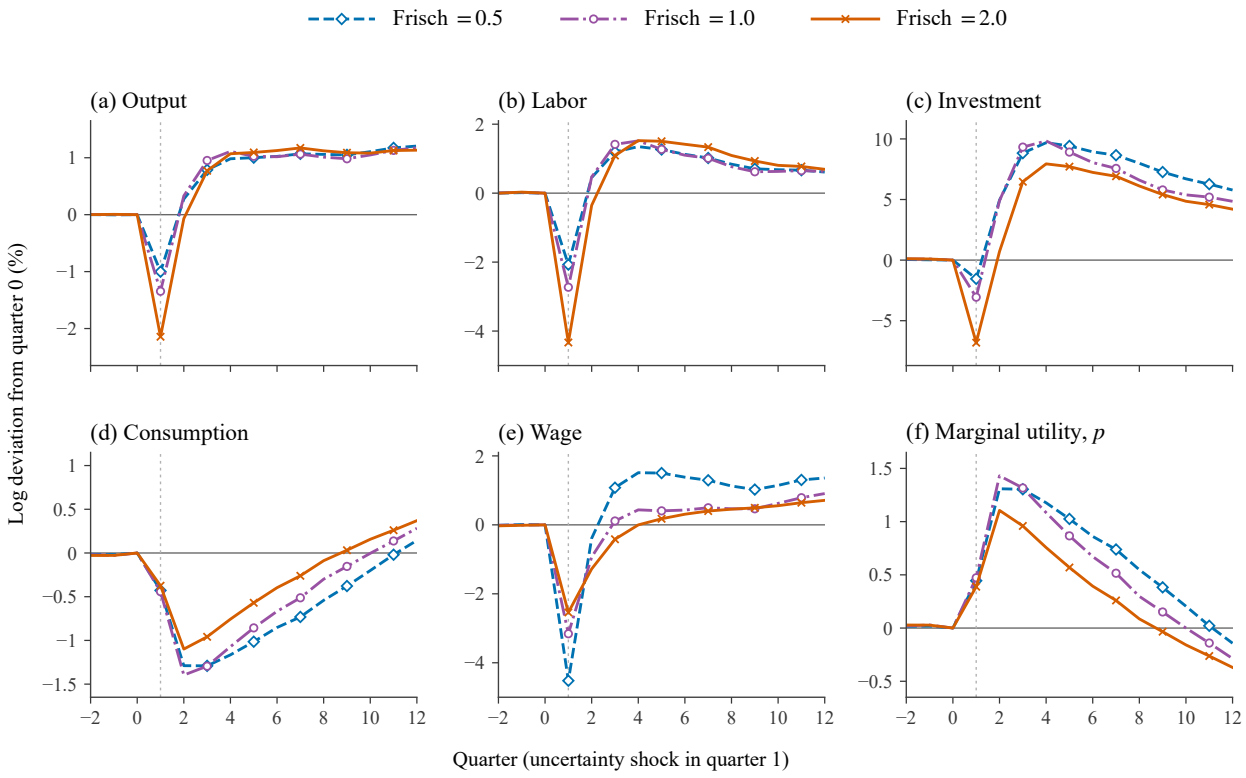


Figure 8: Uncertainty-shock responses as the Frisch elasticity varies

Notes: The uncertainty shock occurs in quarter 1. Responses are $100\log(\overline{X}_t/\overline{X}_0)$, where $\overline{X}_t$ is the arithmetic mean across 2,000 simulated economies within each specification. Quarter 0 is the pre-shock reference period. The figure uses the same panel order and time window as Figure 6; p denotes the marginal utility of consumption. All cases use separable preferences with log utility from consumption and Frisch elasticities 0.5, 1, and 2. Paths are normalized by their own pre-shock means.

Calibration of the Frisch elasticity. Although a smaller elasticity strengthens the wage response that supports employment after an uncertainty shock, it also dampens unconditional labor fluctuations. Table 11 reports labor business-cycle statistics from the final, outer-20 simulation of each case. The standard deviation of labor rises from 0.868% at an elasticity of 0.5 to 1.050% at 1 and 1.601% at 2. The labor-to-output volatility ratio increases from 0.665 to 0.747 and 0.989, and the correlation of labor with output also rises. The corresponding U.S. statistics reported by Bloom et al. (2018, Table 7) are 2.0%, 1.3, and 0.9. A larger elasticity therefore moves all three labor moments toward the data in this comparison, although the elasticity of 2 still understates them.

Table 11: Labor business-cycle statistics across Frisch elasticities

Specification	$\mathrm{sd}(N)$ (%)	$\mathrm{sd}(N)/\mathrm{sd}(Y)$	$\mathrm{corr}(N, Y)$
Frisch $= 0.5$	0.868	0.665	0.544
Frisch $= 1$	1.050	0.747	0.639
Frisch $= 2$	1.601	0.989	0.744
U.S. data	2.0	1.3	0.9

Notes: Model statistics use the final simulation at outer iteration 20, with 5,000 quarters and the first 500 discarded. Each series is expressed in logarithms and HP-filtered with $\lambda = 1600$; the cyclical component is multiplied by 100. Standard deviations use the sample definition. U.S. moments are from Bloom et al. (2018, Table 7), for 1972Q1–2010Q4; labor is total nonfarm business-sector hours. Data entries retain the precision reported in that source.

This calibration consideration is consistent with Bloom et al. (2018), who emphasize that their high assumed Frisch elasticity helps generate realistic hours volatility relative to output. I retain a finite elasticity to preserve the wage channel, while choosing a relatively elastic specification within the range supported by literature.

C.3 Sensitivity to Consumption–Hours Complementarity

This subsection extends the comparison in Section 5.2 to five values of consumption–hours complementarity, $\bar{\kappa} \in \{0, 0.25, 0.5, 0.75, 1\}$. All specifications target $\bar{\zeta} = 1$ and a Frisch elasticity of 2 at the reference allocation, using the calibration in Appendix D.1. The endpoints are separable and GHH preferences, and the three interior values use quasi-separable preferences. The identity $\bar{\zeta} = \bar{\kappa} + \gamma$ implies $\gamma = 1 - \bar{\kappa}$: increasing complementarity also weakens the wealth effect on labor supply. The exercise therefore traces a calibrated family of preferences, rather than varying complementarity while holding the wealth effect fixed.

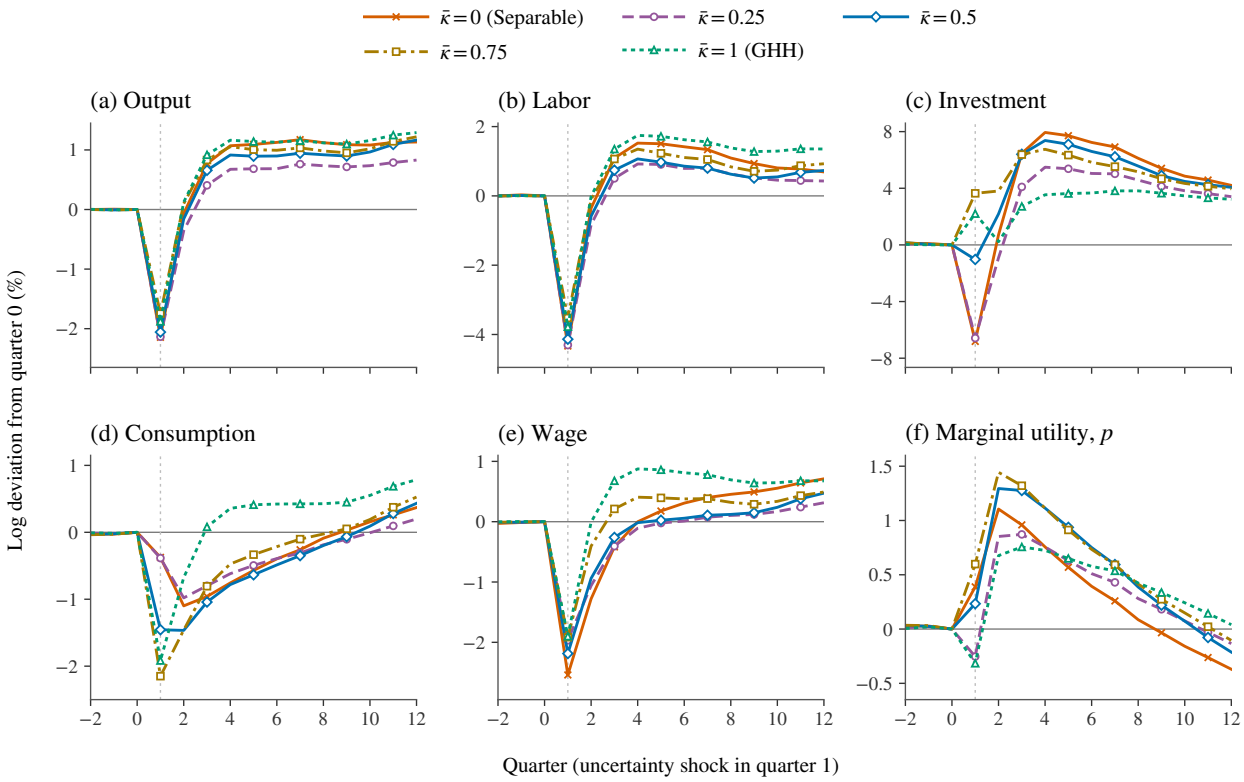


Figure 9: Uncertainty-shock responses as consumption–hours complementarity varies

Notes: The uncertainty shock occurs in quarter 1. Responses are $100\log(\overline{X}_t/\overline{X}_0)$, where $\overline{X}_t$ is the arithmetic mean across 2,000 simulated economies within each specification. Quarter 0 is the pre-shock reference period. The figure uses the same panel order and time window as Figure 6; p denotes the marginal utility of consumption. All cases target $\bar{\zeta} = 1$ and a Frisch elasticity of 2. Paths are normalized by their own pre-shock means.

Figure 9 shows that output, hours, consumption, and the real wage fall on impact in all five cases. Investment is more sensitive to the preference specification: its impact response is negative at $\bar{\kappa} = 0$, 0.25, and 0.5, but positive at 0.75 and 1. Thus, the simultaneous decline in output, consumption, and investment found for the intermediate specification in the main text remains a feature of part of this family, rather than a prediction for every degree of complementarity.

Table 12: Impact responses across complementarity calibrations

$\bar{\kappa}$	γ	Output	Labor	Investment	Consumption	Wage	p
0	1	−2.14	−4.33	−6.81	−0.37	−2.54	0.39
0.25	0.75	−2.14	−4.31	−6.58	−0.39	−1.97	−0.25
0.5	0.5	−2.06	−4.14	−1.03	−1.46	−2.19	0.23
0.75	0.25	−1.75	−3.52	3.64	−2.15	−1.91	0.60
1	0	−1.88	−3.78	2.21	−1.92	−1.90	−0.32

Notes: Entries for quantities and prices are impact log deviations in percent from each specification's pre-shock cross-economy mean, as in Figure 9. The wealth-effect parameter is $\gamma = 1 - \bar{\kappa}$.

Non-monotonic responses. Table 12 makes clear that the responses are not ordered monotonically by $\bar{\kappa}$. For example, raising $\bar{\kappa}$ from 0.5 to 0.75 changes investment's impact response from −1.03% to 3.64%, but a further increase to 1 lowers it to 2.21%. Consumption falls by 1.46%, 2.15%, and 1.92% in these three cases. The contractions in output and hours also become larger between 0.75 and 1. Neither the wage response nor the response of marginal utility follows a monotonic ordering over the five calibrations.

D A Unified Representation of Separable and GHH Preferences

This appendix relates the separable, GHH, and quasi-separable preferences used in the quantitative analysis through the framework of Bilbiie et al. (2025). The key advantage of this representation is that it separates three distinct properties of preferences: the curvature of the marginal utility of consumption, consumption–hours complementarity, and the wealth effect on labor supply.

Consider a general instantaneous utility function

$$U(C, N),$$

where C denotes consumption and N denotes hours worked. Following Bilbiie et al. (2025), define the local curvature of the marginal utility of consumption as

$$\zeta(C, N) \equiv -\left.\frac{\partial \log U_C}{\partial \log C}\right|_N = -\frac{C U_{CC}}{U_C}. \tag{51}$$

I measure consumption–hours complementarity by

$$\kappa(C, N) \equiv -\left.\frac{\partial \log(-U_N)}{\partial \log C}\right|_N = -\frac{C U_{CN}}{U_N}. \tag{52}$$

Because $U_N < 0$, positive Edgeworth complementarity, $U_{CN} > 0$, implies $\kappa > 0$. Finally, define the strength of the wealth effect on labor supply by the response of the marginal rate of substitution

to consumption:

$$\gamma(C,N) \equiv \left.\frac{\partial \log\left(-U_N/U_C\right)}{\partial \log C}\right|_N. \tag{53}$$

Combining (51)–(53) gives

$$\begin{aligned}\gamma(C,N) &= \frac{CU_{CN}}{U_N} - \frac{CU_{CC}}{U_C} \\ &= \zeta(C,N) - \kappa(C,N),\end{aligned} \tag{54}$$

or, equivalently,

$$\boxed{\zeta(C,N) = \kappa(C,N) + \gamma(C,N).} \tag{55}$$

This decomposition is useful because it makes clear that the curvature of marginal utility need not be identified with either consumption–hours complementarity or the wealth effect individually. Instead, ζ reflects both forces.

Separable preferences. Consider first the standard separable specification

$$U^{\text{SEP}}(C,N) = \frac{C^{1-\eta}}{1-\eta} - \chi\frac{N^{1+\phi}}{1+\phi}. \tag{56}$$

Since $U_{CN} = 0$, separability implies

$$\kappa^{\text{SEP}} = 0. \tag{57}$$

Moreover,

$$-\frac{CU_{CC}}{U_C} = \eta,$$

so that

$$\zeta^{\text{SEP}} = \gamma^{\text{SEP}} = \eta. \tag{58}$$

Thus, under separable preferences, all of the curvature in the marginal utility of consumption appears as a wealth effect on labor supply. There is no consumption–hours complementarity.

GHH preferences. At the opposite endpoint, consider GHH preferences,

$$U^{\text{GHH}}(C,N) = \frac{X(C,N)^{1-\eta}}{1-\eta}, \qquad X(C,N) \equiv C - \chi\frac{N^{1+\phi}}{1+\phi}. \tag{59}$$

The marginal utilities are

$$U_C = X^{-\eta}, \qquad U_N = -\chi N^{\phi} X^{-\eta}.$$

It follows that

$$\zeta^{\mathrm{GHH}}(C,N) = \eta \frac{C}{X(C,N)} \tag{60}$$

and

$$\kappa^{\mathrm{GHH}}(C,N) = \eta \frac{C}{X(C,N)}. \tag{61}$$

Hence,

$$\boxed{\zeta^{\mathrm{GHH}} = \kappa^{\mathrm{GHH}}, \qquad \gamma^{\mathrm{GHH}} = 0.} \tag{62}$$

GHH preferences therefore represent the other endpoint: all of the curvature of the marginal utility of consumption is associated with consumption–hours complementarity, while the wealth effect on labor supply is eliminated completely.

Notice that the primitive curvature parameter η in the GHH utility function is not itself equal to ζ. Rather, the relevant local curvature is

$$\zeta^{\mathrm{GHH}}(C,N) = \eta \frac{C}{X(C,N)}.$$

This distinction is important when comparing GHH and separable preferences while holding ζ fixed.

Quasi-separable GHH–CRRA preferences. The quasi-separable specification of Bilbiie et al. (2025) nests these two polar cases while allowing complementarity and the wealth effect to be controlled separately:

$$U^{\mathrm{QS}}(C,N) = \frac{1}{1-\frac{\xi}{1-\gamma}} \left[\frac{C^{1-\gamma}}{1-\gamma} - \chi \frac{N^{1+\phi}}{1+\phi} \right]^{1-\frac{\xi}{1-\gamma}}. \tag{63}$$

Define

$$A(C,N) \equiv \frac{C^{1-\gamma}}{1-\gamma} - \chi \frac{N^{1+\phi}}{1+\phi}, \qquad a \equiv \frac{\xi}{1-\gamma}. \tag{64}$$

The marginal utilities are

$$U_C = C^{-\gamma} A^{-a}, \qquad U_N = -\chi N^{\phi} A^{-a}. \tag{65}$$

Therefore,

$$\zeta^{\mathrm{QS}}(C,N) = \gamma + a \frac{C^{1-\gamma}}{A(C,N)}, \tag{66}$$

$$\kappa^{\mathrm{QS}}(C,N) = a \frac{C^{1-\gamma}}{A(C,N)} = \frac{\xi}{1-\gamma} \frac{C^{1-\gamma}}{A(C,N)}. \tag{67}$$

Consequently,

$$\boxed{\zeta^{\mathrm{QS}}(C,N) = \kappa^{\mathrm{QS}}(C,N) + \gamma.} \tag{68}$$

The two benchmark specifications arise as limiting cases. When complementarity disappears, $\kappa = 0$, the quasi-separable specification converges to separable CRRA preferences. At the other endpoint, setting $\gamma = 0$ eliminates the wealth effect and gives

$$A(C, N) = C - \chi \frac{N^{1+\phi}}{1+\phi},$$

so that (63) reduces exactly to the GHH form.

D.1 Calibration with a Common Consumption Curvature

To isolate the role of consumption–hours complementarity, I hold the curvature of the marginal utility of consumption fixed across preference specifications. Specifically, I impose

$$\bar{\zeta} = 1, \tag{69}$$

where a bar denotes evaluation at the deterministic steady state. I then vary $\bar{\kappa}$ and adjust the wealth-effect parameter according to

$$\boxed{\gamma = \bar{\zeta} - \bar{\kappa} = 1 - \bar{\kappa}.} \tag{70}$$

Hence, increasing complementarity while holding ζ fixed necessarily reduces the wealth effect.

For a target with $\gamma \neq 1$, define

$$\bar{A} = \frac{\bar{C}^{1-\gamma}}{1-\gamma} - \chi \frac{\bar{N}^{1+\phi}}{1+\phi}. \tag{71}$$

Equation (67) implies that ξ is chosen according to

$$\boxed{\xi = \bar{\kappa}(1-\gamma)\frac{\bar{A}}{\bar{C}^{1-\gamma}}.} \tag{72}$$

Thus, the calibration proceeds in two steps. First, for a given target $(\bar{\zeta}, \bar{\kappa})$, γ is determined from (70). Second, ξ is chosen from (72) so that the desired degree of complementarity is attained at the deterministic steady state.

When the same steady-state allocation $(\bar{C}, \bar{N})$ and wage $\bar{w}$ are imposed across preference specifications, the scale of labor disutility is correspondingly adjusted using the steady-state intratemporal condition:

$$\chi = \frac{\bar{w}}{\bar{C}^{\gamma}\bar{N}^{\phi}}. \tag{73}$$

This ensures that differences across specifications reflect complementarity and curvature rather than differences in the deterministic steady state.

I consider three cases.

Separable: $\bar{\kappa} = 0$**.** With $\bar{\zeta} = 1$, equation (70) gives

$$\gamma = 1.$$

Thus,

$$(\bar{\zeta}, \bar{\kappa}, \gamma) = (1, 0, 1).$$

This corresponds to the log-separable specification

$$U^{\text{SEP}}(C, N) = \log C - \chi \frac{N^{1+\phi}}{1+\phi}. \tag{74}$$

Formally, this is the limiting case of the quasi-separable family as $\gamma \to 1$ and $\xi \to 0$. All consumption curvature operates through the wealth effect, and there is no consumption–hours complementarity.

Quasi-separable: $\bar{\kappa} = 0.5$**.** For the intermediate specification, I impose

$$\bar{\kappa} = 0.5.$$

Holding $\bar{\zeta} = 1$ implies

$$\gamma = 1 - 0.5 = 0.5. \tag{75}$$

Hence,

$$(\bar{\zeta}, \bar{\kappa}, \gamma) = (1, 0.5, 0.5).$$

The corresponding steady-state aggregator is

$$\bar{A}_{0.5} = 2\bar{C}^{1/2} - \chi \frac{\bar{N}^{1+\phi}}{1+\phi},$$

and (72) gives

$$\boxed{\xi_{0.5} = \frac{1}{4} \frac{\bar{A}_{0.5}}{\bar{C}^{1/2}}.} \tag{76}$$

This specification therefore assigns one half of the common consumption curvature to complementarity and the other half to the wealth effect.

GHH: $\bar{\kappa} = 1$**.** Finally, imposing

$$\bar{\kappa} = 1$$

together with $\bar{\zeta} = 1$ requires

$$\gamma = 0.$$

Therefore,

$$(\bar{\zeta}, \bar{\kappa}, \gamma) = (1, 1, 0),$$

and the quasi-separable utility reduces to GHH. Let

$$\bar{X} = \bar{C} - \chi \frac{\bar{N}^{1+\phi}}{1+\phi}.$$

Since under GHH

$$\bar{\zeta} = \bar{\kappa} = \xi \frac{\bar{C}}{\bar{X}},$$

the curvature parameter is chosen as

$$\boxed{\xi^{\text{GHH}} = \frac{\bar{X}}{\bar{C}}.} \tag{77}$$

This calibration delivers $\bar{\zeta} = \bar{\kappa} = 1$ while eliminating the wealth effect completely.

The three calibrations can therefore be summarized as

	$\bar{\zeta}$	$\bar{\kappa}$	γ
Separable	1	0	1
Quasi-separable	1	0.5	0.5
GHH	1	1	0

(78)

Moving from separable to GHH preferences can thus be interpreted as progressively replacing the wealth-effect component of marginal-utility curvature with consumption–hours complementarity, while holding the total curvature $\bar{\zeta}$ fixed.

It is important that ζ and κ in the quasi-separable specification are local objects. Although the calibration imposes the desired values at the deterministic steady state, both generally vary with (C, N) away from the steady state. The wealth-effect parameter γ, in contrast, is constant. Nevertheless, the decomposition

$$\zeta(C, N) = \kappa(C, N) + \gamma$$

continues to hold pointwise along the entire equilibrium path.